



Lecture Session 11

1. Forces acting on the ocean;
2. Earth Rotation: The Coriolis effect;
3. Driving mechanisms of ocean currents.

Icebreaker discussion. In groups ...

1. Each group will identify three anthropogenic influences on the marine environment
2. Each group will take a turn sharing their information with the class and writing them on a board or in the zoom chat.
3. Discuss ways to categorize the anthropogenic influences into groups based on similarities

An aerial photograph of a tropical coastline. A narrow, white sandy beach runs diagonally from the bottom left towards the top right. To the left of the beach is a shallow lagoon with clear, turquoise water, revealing a complex coral reef structure with various ridges and channels. To the right of the beach is a deeper, darker blue ocean. The overall scene is serene and natural, with no visible signs of pollution.

Marine Pollution

Marine pollutants may be natural or human-generated

- **Marine pollution** – affects the physical, chemical, or biological environment
- Pollution can be **chronic** or **acute**
- Pollutants interfere with mechanical or biological processes in an organism. The effects will depend on the pollutant's
 - Quantity
 - Toxicity
 - Persistence



Tom Garrison

Icebreaker discussion 2. In groups ...

1. Each group will identify one natural pollution source
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Few examples: Volcanic activity, natural eutrophication, natural oil seeps, natural harmful algal blooms, intense changes to oceanographic conditions (Ex: intense upwelling events).

Main sources of human-generated marine pollutants

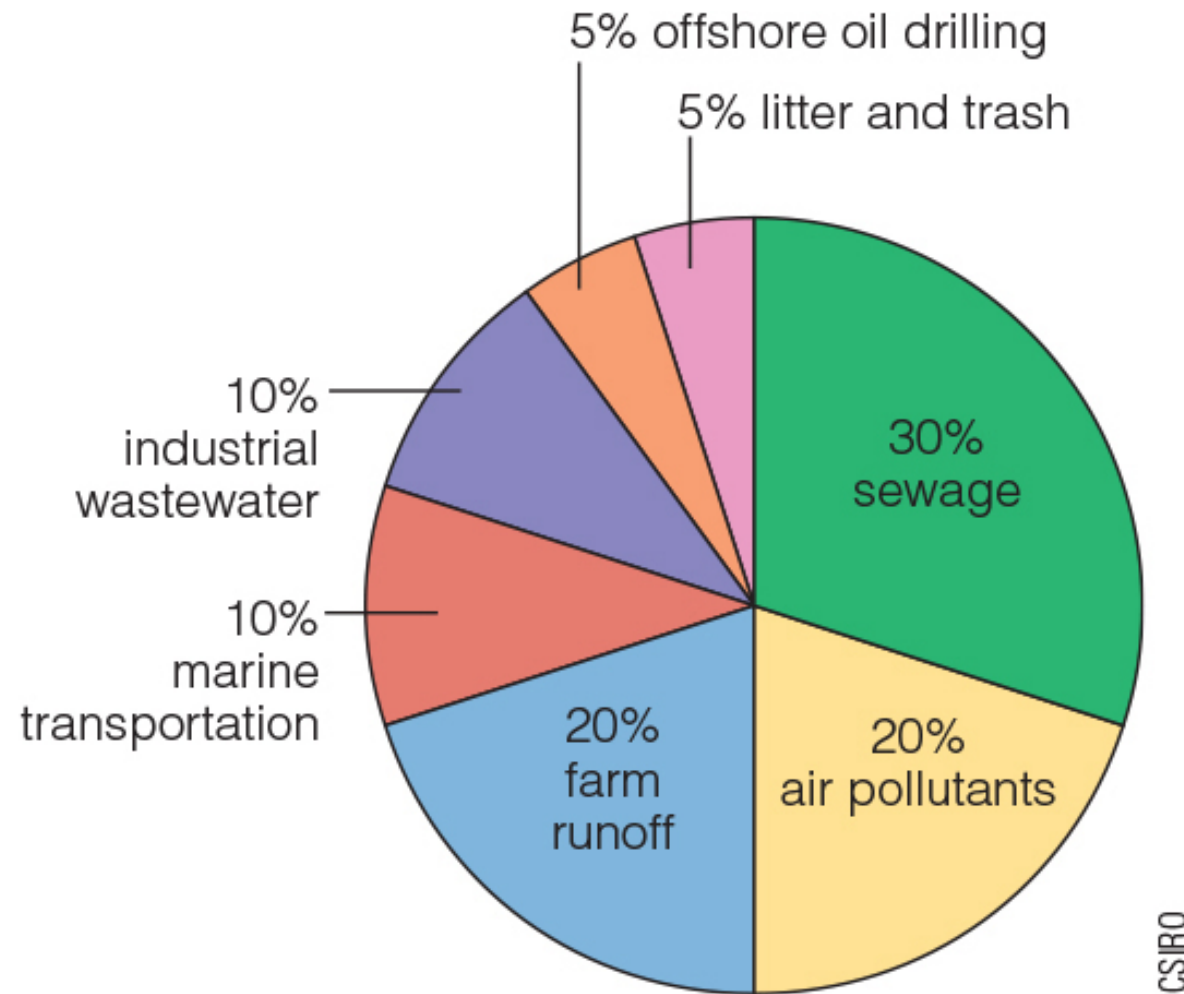


Figure 18.2 Sources of marine pollution.

An aerial photograph of a coastline. The ocean is a deep, dark blue. The shoreline is irregular, with a mix of green and brownish-yellow patches, possibly indicating vegetation or a spill. A large, irregular, light-colored area, likely an oil spill, is visible in the upper right quadrant, extending from the land into the water. The text "Oil in the ocean" is overlaid in white, sans-serif font in the center of the image.

Oil in the ocean

Natural oil seeps is the major source of crude oil to the marine environment. However, its chronic, constant, and small spread out nature makes it less “impacting”.

- Natural oil seeps: crude oil
- Human-caused oil in the ocean
 - Crude oil
 - Refined oil

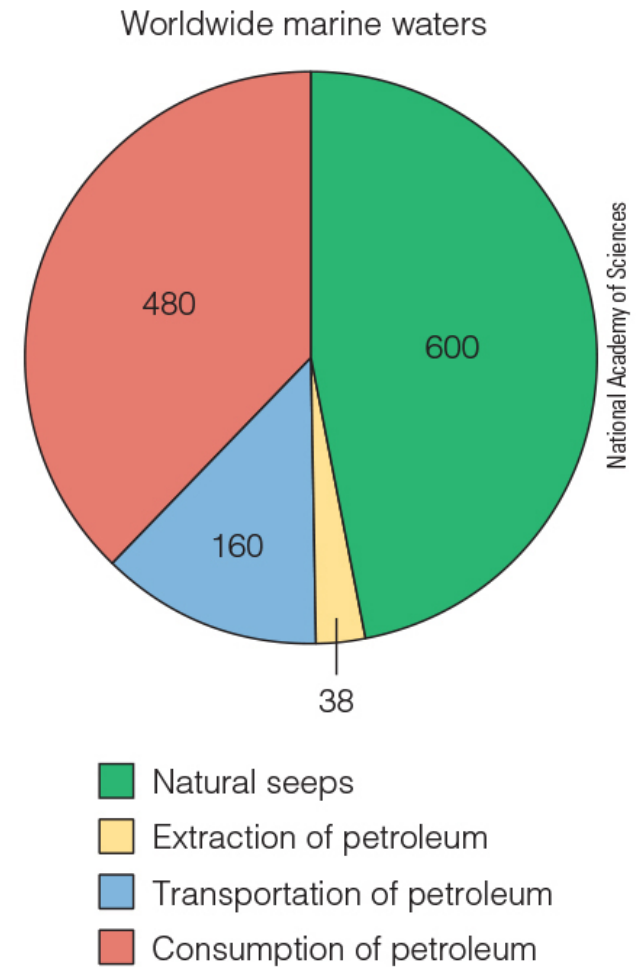


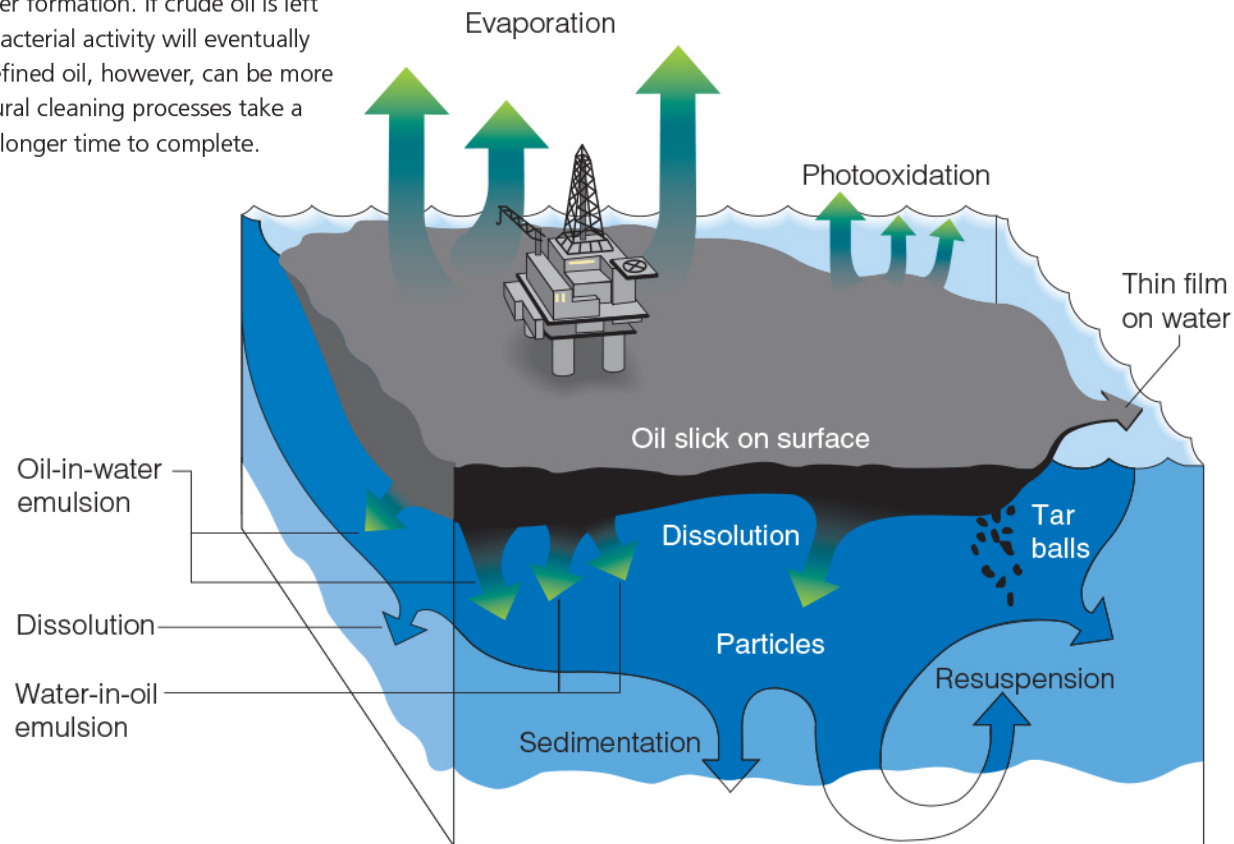
Figure 18.3 Average worldwide annual releases of petroleum by source (1990–1999), in thousands of metric tons per year.

Oil Spills

Acute pollution source with long-lasting environmental and economic impacts

Figure 18.6 The fate of oil spilled at sea.

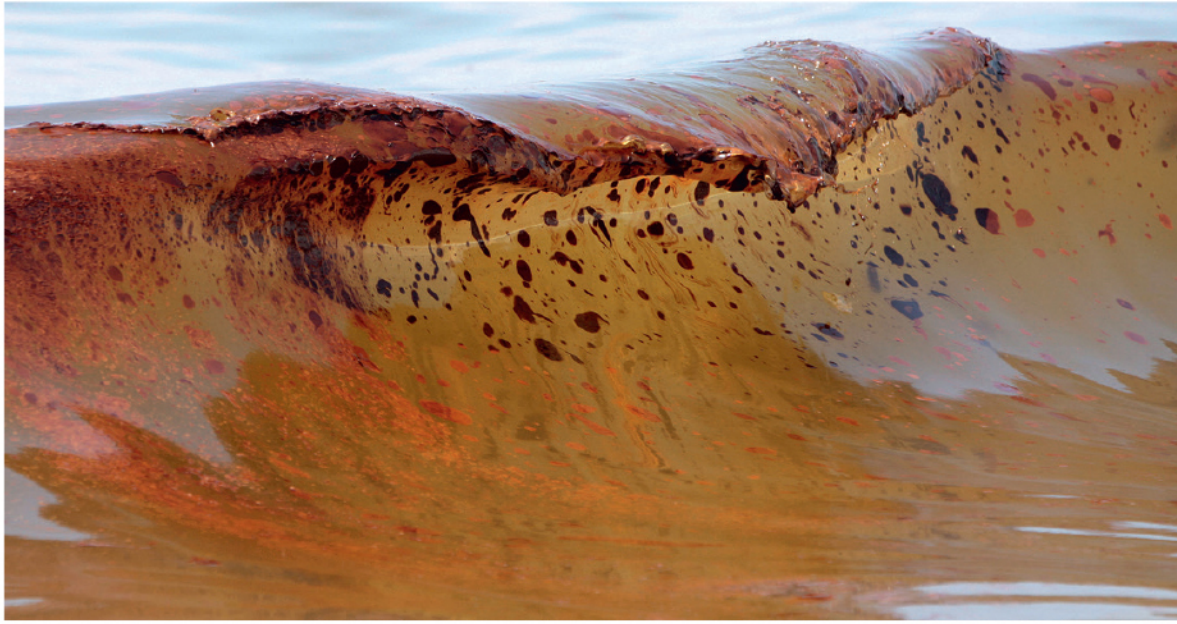
Smaller molecules evaporate or dissolve into the water beneath the slick. Within a few days, water motion coalesces the oil into tar balls and semisolid emulsions of water in oil and oil in water. Tar balls and emulsions may persist for months after formation. If crude oil is left undisturbed, bacterial activity will eventually consume it. Refined oil, however, can be more toxic, and natural cleaning processes take a proportionally longer time to complete.



U.S. Coast Guard photo

Figure 18.8 Deepwater Horizon moments before sinking.

AP Images/Dave Martin



- a** Congealed crude oil washes up on Orange Beach, Alabama, about 145 kilometers (90 miles) from the site of the *Deepwater Horizon* spill.

- b** An oil-soaked brown pelican being rescued from the fouled marshlands of Louisiana nearly a month and a half after the *Deepwater Horizon* spill.

UPI/Alamy Stock Photo

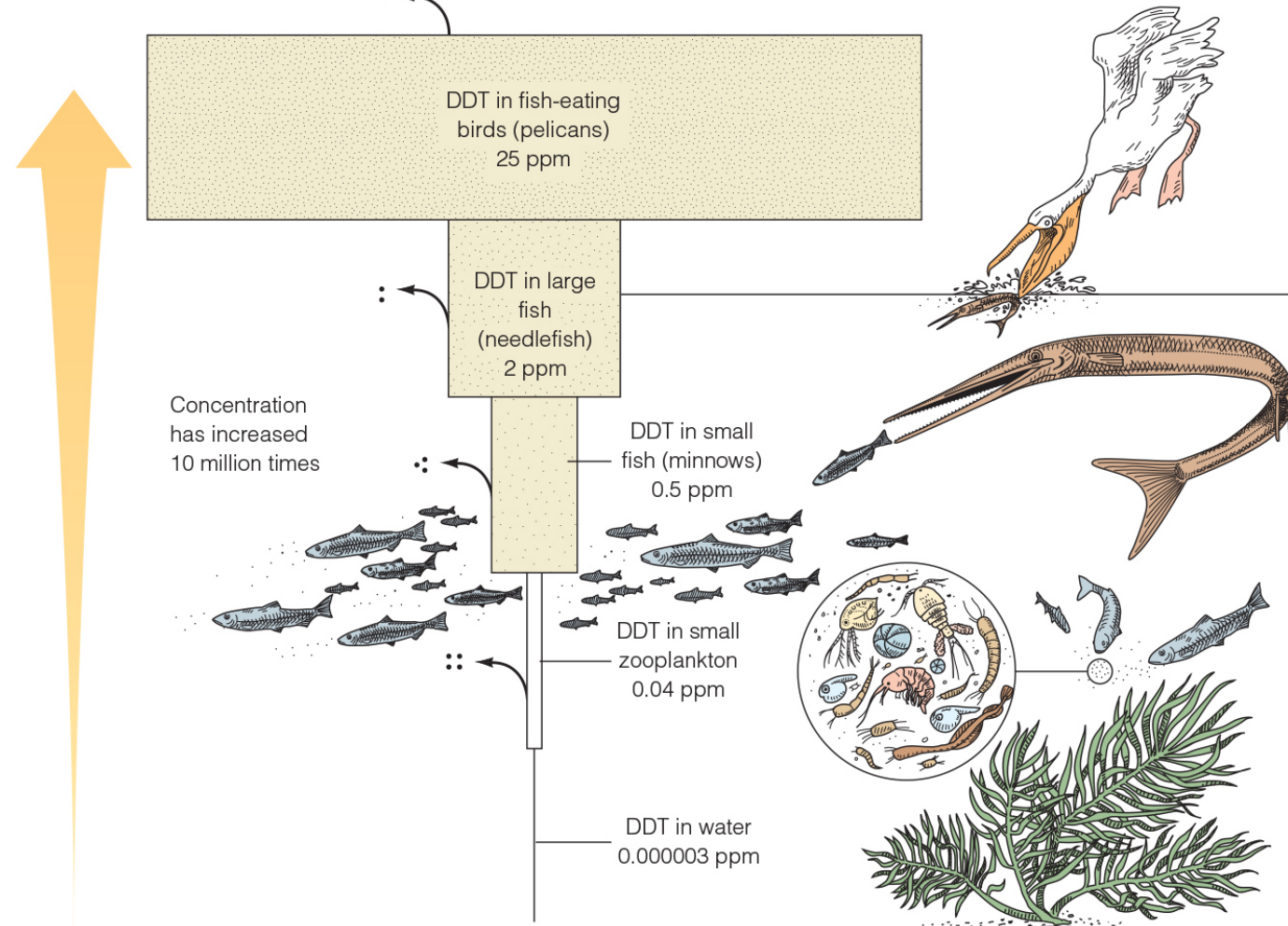


Figure 18.5 Crude oil spills.

An aerial photograph of a coastline. The top half of the image shows a dark blue ocean with some white foam from waves breaking. The bottom half shows a land area with a mix of green and brown, suggesting vegetation and possibly some industrial or agricultural areas. The text "Bioaccumulation of toxic chemicals" is overlaid in white, centered horizontally and slightly above the middle vertically.

Bioaccumulation of toxic chemicals

Heavy metals 🤘 (Ex: mercury, lead), pesticides, and plastic can bioaccumulate (biomagnification) along the food chain



An aerial photograph of a coastline. The top half of the image shows deep blue ocean water. A narrow, light-colored sandy beach runs diagonally across the middle. Below the beach is a shallow, turquoise-colored lagoon or reef flat. The bottom half of the image shows a vast, flat, greenish-brown landscape, likely a salt flat or a coastal plain. The text "Plastic (maybe the worst of all)" is overlaid in white, centered in the upper half of the image.

Plastic (maybe the worst of all)

Toxic plastic chemical components can also bioaccumulate. Plastic has been released in the marine environment over several decades (chronic pollutant) with no regulation so far. It can break down into very small pieces (pallets) impacting all sorts of organisms.

A LIFETIME OF PLASTIC

The first plastics made from fossil fuels are just over a century old. They came into widespread use after World War II and are found today in everything from cars to medical devices to food packaging. Their useful lifetime varies. Once disposed of, they break down into smaller fragments that linger for centuries.

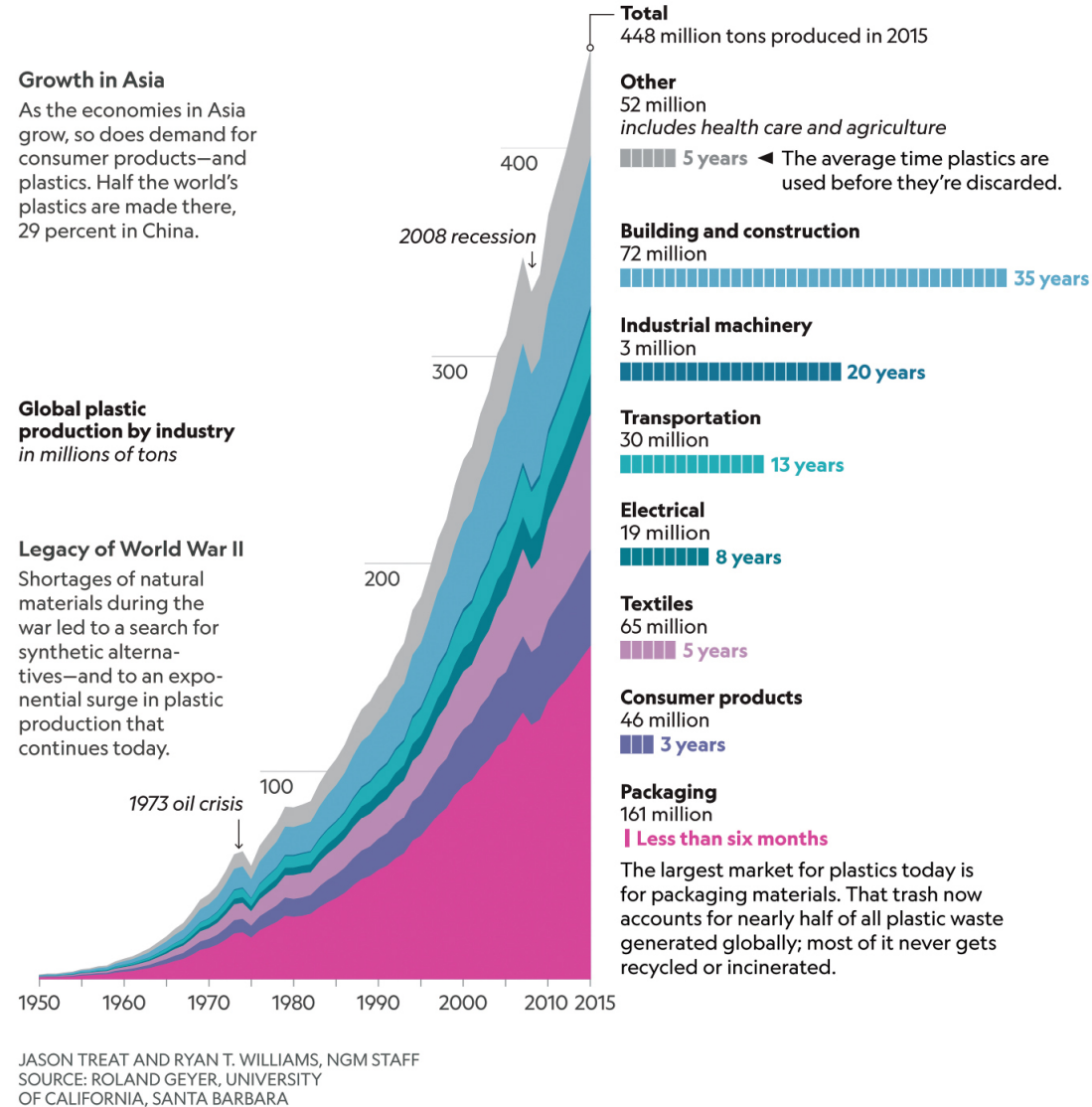


Figure 18.15 The amount of plastic produced over the last half century has risen nearly exponentially.

It impacts from large organisms



Figure 18.16 A bottlenose dolphin swims through a sea of marine debris dominated by plastic waste.



a Sea lions (seen here) and seals die by the hundreds each year after becoming entangled in plastic debris, especially discarded and broken fishing nets.



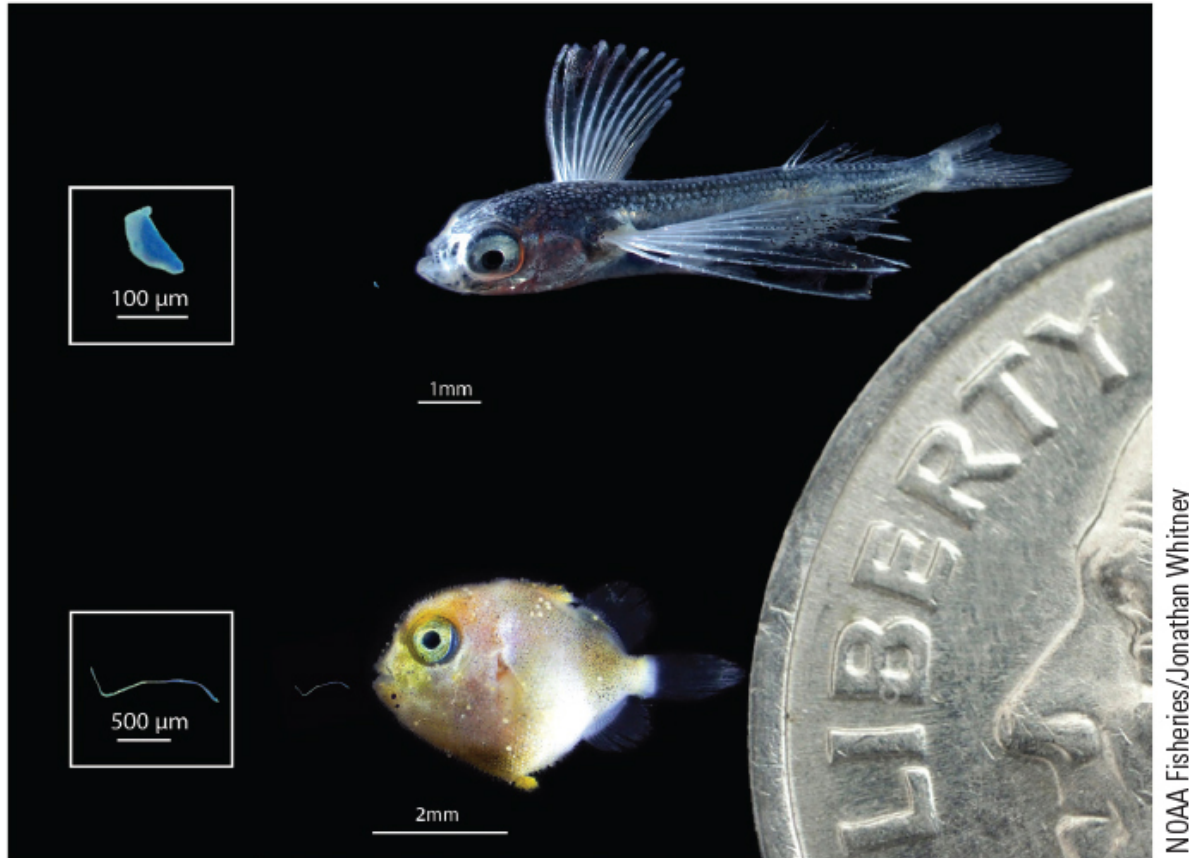
b A decaying albatross reveals the cause of its death. Plastic trash (mistaken for food) has blocked its digestive system. On Midway Atoll, about 40% of albatrosses die in this way.



c Some species of sea turtles can mistake floating plastic bags for their natural food source, jellies.

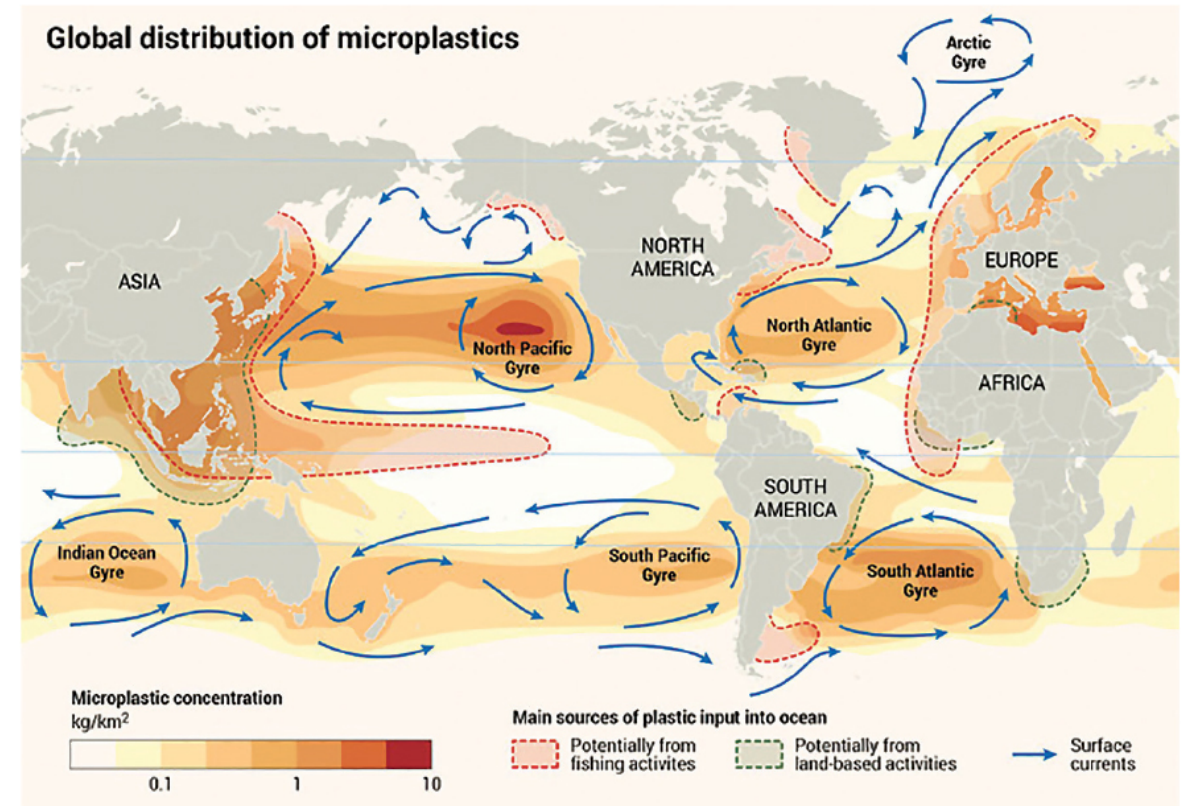
Figure 18.17 Discarded plastic can pose direct or indirect dangers to marine life.

To the smallest everywhere in the ocean



- b** Larval fish have been found to ingest toxin-laden microplastics at a stage in their life history that is critical to their development. The dime on the right side of the image is shown for scale.

Ocean circulation spreads plastics around



- c** The global distribution of microplastics. Note their concentration and distribution in relationship to the major gyres and that the primary sources from fishing activities and land-based activities overlap in some areas and are different in others.